A Low-Noise Multiresolution High-Dynamic Ultra-Broad-Band Time-Domain EMI Measurement System

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***Abstract*—A time-domain electromagnetic interference (EMI) measurement system allows to reduce measurement time by several orders of magnitude in comparison with conventional systems. This will considerably reduce the costs of compliance tests.**

In this paper, a novel multiresolution time-domain EMI measurement system is presented. The multiresolution time-domain EMI measurement system exhibits several channels with limiters, amplifiers, and analog-to-digital converters. The amplitude range of the signal is subdivided into several intervals. In each interval the analog-to-digital conversion is performed with an amplitude discretization proportional to the width of the amplitude interval. The signal-to-noise ratio of the multiresolution time-domain EMI measurement system compared with a conventional time-domain measurement system for transient signals is enhanced by at least 50 dB. It is shown that the multiresolution time-domain EMI measurement system has a higher overload factor than conventional systems working frequency domain. The noise floor is 5 dB lower than the noise floor of a conventional EMI receiver. The multiresolution time-domain EMI measurement system shows a higher 1-dB compression point than conventional EMI receivers. Measurements were performed in the frequency range from 30 MHz to 1 GHz and compared with results obtained by an EMI receiver.

***Index Terms*—Analog-to-digital converter (ADC), electromagnetic compatibility (EMC), electromagnetic interference (EMI), low noise, time domain.**

I. Introduction

TRADITIONALLY electromagnetic interference (EMI) measurements are performed with conventional analog EMI receivers operated in the frequency domain. Measurements in the frequency domain take long measurement time, up to several hours for a single frequency scan. EMI receivers use a pre-selection to obtain the required dynamic range of 36 dB according to the standard CISPR 16-1 [2]. The standards are published by the International Special Committee on Radio Interference (CISPR), which was established in 1934. An efficient way to reduce the measurement time is to use a time-domain EMI measurement system [3]. The measurement time can be reduced by several orders of magnitude via this system. A time-domain EMI measurement system for pre-compliance measurements [1] shows a limited dynamic range given by the resolution of the used analog-to-digital converter (ADC).

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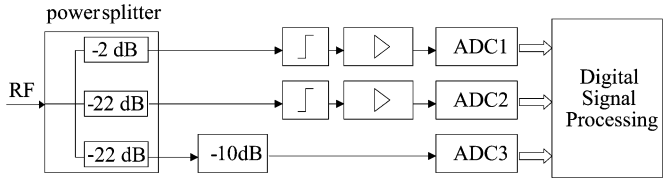


Fig. 1. Low-noise multiresolution time-domain EMI measurement system.

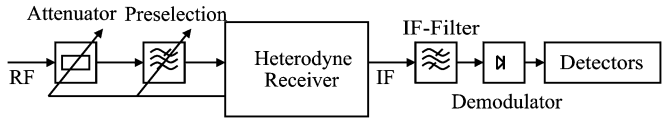


Fig. 2. Conventional full-compliance EMI receiver.

Today, high-speed ADCs with a sampling rate of 5 GS/s are available with a resolution of only 8 bit. Thus, for stationary signals, the signal-to-noise ratio (SNR) is approximately 50 dB. In this case, the time-domain EMI measurement system already fulfills the CISPR 16-1. However, for the transient signal, the SNR is usually below 10 dB. The operation of a multiresolution time-domain EMI measurement system has been presented in [4]. In the following, a novel low-noise multiresolution time-domain EMI measurement system with an improved dynamic range and low input reflection is presented.

In Fig. 1, the block diagram is shown. The input signal is distributed by a power splitter into three channels. Each channel consists of a limiter, a low-noise amplifier, and an ADC. While the first channel digitizes the amplitude range from 0 to 1.8 mV, the third channel digitizes the amplitude range from 0 to 10 V. The second channel is used to digitize the intermediate amplitude range from 0 to 200 mV. With optional digitally controlled p-i-n attenuators, the amplitude range can be adapted to the input signal to obtain a maximum SNR. The signal is recorded at all three channels simultaneously. A signal digitized in high resolution is reconstructed by extracting each sampled value from the ADC where the signal shows the maximum nonclipped value. The reconstructed signal is processed via a short-time fast Fourier transform to a spectrogram. From the obtained time-dependent spectrum, the peak, quasi-peak, average, and rms value are calculated [5].

II. Dynamic Performance of EMI Receivers

Here, the dynamic performance of conventional EMI receivers is described. The block diagram of an EMI receiver is depicted in Fig. 2. The input signal is attenuated with a

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controlled attenuator. The pre-selection of the EMI receiver typically consists of several selectable bandpass filters, which are tunable within a certain frequency range. After pre-selection, the signal is downconverted by a heterodyne receiver to the IF. Selectable IF filters according to the standard CISPR 16-1 [2] are applied to the IF signal. The IF signal is demodulated and evaluated by the different selectable detectors peak, quasi-peak, average, and rms.

A. Dynamic Performance

The dynamic performance of EMI receivers is described by several characteristics [2]. These characteristics are compared with the dynamic characteristics of the multiresolution time-domain EMI measurement system.

1) Spurious-Free Dynamic Range (SFDR): The SFDR) of an EMI receiver is limited by nonlinear effects and other spurious responses. The SFDR is described for stationary signals as follows:

$$\mathrm{SFDR}_{\mathrm{dB}} = \max(S_{\mathrm{dB}}[f]) - \max(H_{\mathrm{dB}}[f]) \tag{1}$$

where $S[f]$ is the power spectrum of the input signal applied to the system and $H[f]$ are spurious signals that may be harmonics mixing products. $[f]$ means the set of frequency that are selected during the stepped frequency scan.

For a transient broad-band signal, the SFDR is determined by a notch filter test [2]. A pulse generator with a uniform spectrum within the measurement band is connected via a notch filter to the measurement system. The measured attenuation at the blocking frequency is the SFDR for the transient signal. This procedure is described in CISPR 16-1. CISPR 16-1 demands a dynamic range of 40 dB for the stationary signal and 36 dB for the transient signal. The SFDR has to be fulfilled for all detector modes.

2) Dynamic Range: The dynamic range is defined as follows:

$$\mathrm{DR}_{\mathrm{dB}} = S_{\mathrm{max,dB}}[f] - \max\left(N_{\mathrm{dB}}[f]\right) \tag{2}$$

where $S_{\mathrm{max,dB}}[f]$ is the maximum displayable value and $N_{\mathrm{dB}}[f]$ is the noise floor. The achievable dynamic range also depends on the signal type and detector mode.

3) Overload Factor: The overload factor describes the dynamic range that can be processed in the IF section. An EMI receiver must provide a minimum overload factor of 43.5 dB. This overload factor only applies to transient signals. The overload factor is defined as follows:

$$\mathrm{OF}_{\mathrm{dB}} = \max(S_{\mathrm{IF,dB}}) - \max\left(N_{\mathrm{IF,dB}}\right)\big) \tag{3}$$

where $S_{\mathrm{IF,dB}}$ is the maximum signal power that can be processed by the IF section and $N_{\mathrm{IF,dB}}$ is the noise within the IF section. The overload factor is independent of the selected detector mode.

B. Controlled Attenuation

EMI receivers use a selectable attenuation to get an enhanced dynamic range. During a frequency scan, the EMI receiver selects the optimum attenuation to obtain a maximum SNR at the IF stage. One drawback is that, for each selected attenuation level, a calibration has to be performed to obtain accurate measurements. Another disadvantage is that the mechanical switches that are used to select different attenuator levels have limited cycles of switching.

C. Preselection

EMI Receivers use a pre-selective filter to obtain an enhanced dynamic range for transient broad-band signals and narrow-band signals. By the pre-selective filter, only the power of the selected frequency band is applied to the heterodyne receiver. Typically, the pre-selective filter has a bandwidth of several megahertz. The enhancement that applies for the transient broad-band signal is expressed by (4) as follows:

$$D_{i,\mathrm{dB}} = 10\log_{10}\left(\frac{B_s}{B_{\mathrm{pre}}}\right) \tag{4}$$

where B_s is the bandwidth of the system and B_{pre} is the equivalent noise bandwidth of the pre-selection. For the stationary signal, the pre-selection suppresses strong narow-band signals that may contribute to an overload. One drawback of the pre-selection is the complexity of the various filters, which yields to the necessity of calibration. Another disadvantage is the input reflection coefficient of such an EMI receiver. The input is only matched at the selected frequency. Measurements have shown that the EMI receivers show a voltage standing-wave ratio (VSWR) of 2.0 at the selected frequency and a VSWR up to 7.0 at the rest of the frequencies.

This might cause significant errors when nonlinear systems are measured via absorber clamps or mains artificial networks because the emission of nonlinear system at a certain frequency depends on the impedance that is provided to the system at other frequencies.

III. Dynamic of a Time-Domain EMI Measurement System With One ADC

Here, we discuss the dynamic range that can be achieved with a time-domain EMI measurement system when a single signal interval is digitized. We only consider quantization noise that is added to the signal during analog-to-digital conversion. We have to consider the SNR of a single spectra calculated by the short-time fast Fourier transformation [6]. By this way, we can describe the dynamic parameters that can be compared with conventional EMI receivers.

A. SNR

For a general signal, the SNR is given by [7]

$$\mathrm{SNR} = \frac{P_{\mathrm{Signal}}}{P_{\mathrm{Noise}}} \tag{5}$$

while the power of the signal is given by (6) as follows:

$$P_{\mathrm{Signal}} = \sigma_s^2 = \left\langle \Big(x(t) - \langle x(t)\rangle\Big)^2 \right\rangle. \tag{6}$$

The power of the quantization noise is given by (7)

$$P_{\text{Noise}} = \sigma_e^2 = \frac{\triangle^2}{12} \tag{7}$$

with $\triangle$ as the quantization step of the ADC.

1) Window Function: Spectral estimation is performed using the discrete Fourier transform (DFT). The DFT is used to calculate the amplitude spectrum of a signal sample of length N. In addition, the DFT is based upon the assumption that this sample is continued periodically in time. Thus, the time record $x[n]$ needs to be multiplied with a window $w[n]$ of length N to avoid spectral leakage

$$z[n] = x[n]w[n]. \tag{8}$$

For the window function $w[n]$, a Gaussian window is used. In order to get at each discrete frequency value, the same filter response as the IF filter of the conventional EMI receiver, the window needs to be adapted to the IF filter regarding impulse bandwidth and equivalent noise bandwidth. The DFT is applied to each of the data blocks and is defined as follows [8]:

$$Z[k] = \sum_{n=0}^{N-1} z[n] e^{-j(2\pi kn/N)}, \qquad k = 1, \ldots, N. \tag{9}$$

2) Window Function: The digitized signal $\hat{x}[n]$ is the result of the superposition of the original signal $x[n]$ and the quantization noise $e[n]$. Using (8), we obtain

$$z[n] = x[n]w[n] = \hat{x}[n]w[n] + e[n]w[n]. \tag{10}$$

$e[n]$ is described as a statistical signal with constant probability density function in the interval $[-(\triangle/2); +(\triangle/2)]$ according to a midriser quantization [7]. Applying (5), (6), and (8), we obtain an SNR for each calculated spectrum as shown in [4]

$$\text{SNR} = \frac{\frac{1}{N}\sum_{n=1}^{N}\left(\hat{x}[n]w[n] - \frac{1}{N}\sum_{k=1}^{N}\hat{x}[k]w[k]\right)^2}{\frac{1}{N}\sum_{n=1}^{N}\left(e[n]w[n] - \frac{1}{N}\sum_{k=1}^{N}e[k]w[k]\right)^2}. \tag{11}$$

For $e[n] \ll x[n]$, we can replace $\hat{x}[n]$ by $x[n]$ and obtain with (11)

$$\text{SNR} \approx \frac{\frac{1}{N}\sum_{n=1}^{N}\left(x[n]w[n] - \frac{1}{N}\sum_{k=1}^{N}x[k]w[k]\right)^2}{\frac{1}{N}\sum_{n=1}^{N}\left(e[n]w[n] - \frac{1}{N}\sum_{k=1}^{N}e[k]w[k]\right)^2}. \tag{12}$$

3) Instantaneous SNR: A instantaneous SNR is described by the SNR of several consecutive spectra [9]. An example of such an instantaneous SNR for a modeled IF bandwidth of 120 kHz, and an 8-bit ADC is shown in Fig. 3 from [4]. The input signal was obtained by the measurement of a laptop and a drill machine simultaneously.

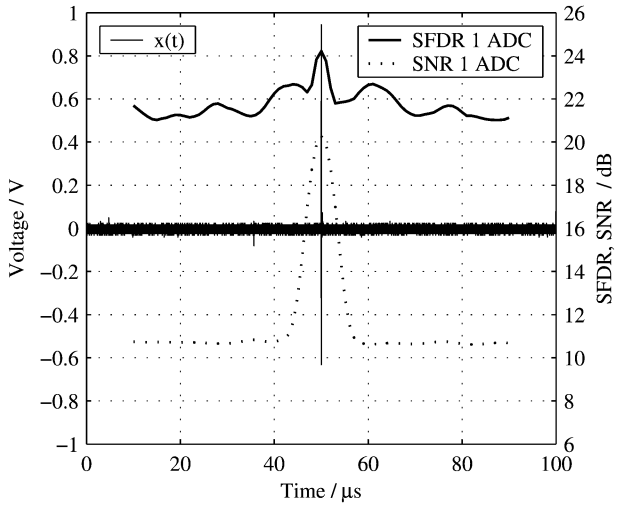


Fig. 3. Instantaneous SNR and dynamic range. From [4].

B. Dynamic Performance

In Section III-A, the SNR of the time-domain EMI measurement system was described. In the following, the interrelationship between the SNR and the dynamic performance of a time-domain EMI measurement system is presented.

The quantization noise is modeled as white Gaussian noise. Therefore, P_{Noise} is equally spread over the frequency range $[0; (f_s/2)]$, where f_s is the sampling frequency. However, P_{Signal} is distributed over the frequency according to its power spectrum $Z[n]$. SFDR_s of a single calculated spectrum is given by

$$\text{SFDR}_s = \frac{\max(Z[n])N}{P_{\text{Noise}}}. \tag{13}$$

An example of the instantaneous SFDR for the transient signal is shown in Fig. 3 from [4]. In the following, the dynamic performance is described by the dynamic parameters of an EMI receiver.

1) SFDR: For the modeling, we have only considered the quantization noise. Measurements have shown that, in the peak detector mode, the noise floor is up to 11 dB higher than the average noise floor [10]. In the rms detector mode, the noise floor is 1.2 dB higher, and in the quasi-peak detector mode, the noise floor is up to 6 dB higher [10]. The standard [2] describes the measurement of the dynamic parameters for the transient signal at very high pulse repetition rates, where indication during the various detector modes to the input signal is almost equal. In the following, we consider only the peak detector mode. The results for the other detector modes can be directly derived from the dynamic parameters in the peak detector mode. From Fig. 3, we obtain a maximum SFDR of 14 dB in the peak detector mode.

2) Dynamic Range: The dynamic range is the difference between the maximum amplitude level that can be applied and the minimum amplitude level that is higher than the noise floor. In this case, the complete amplitude scale is already used by the signal. Thus, the dynamic range is around 14 dB.

3) Overload Factor: For a time-domain EMI measurement system that uses only one ADC, the overload factor is identical with the dynamic range.

C. Stationary Signal

The SNR and dynamic range of the system for transient signal has been investigated above. In the following, the SNR for stationary signal is presented.

1) SNR: The stationary signal has a constant power over the time. Thus, the SNR is constant. With (6) and (7), we obtain a constant SNR of

$$\mathrm{SNR} = 12\frac{\left\langle \left(x(t) - \langle x(t)\rangle\right)^2 \right\rangle}{\triangle^2}. \tag{14}$$

2) Dynamic Range: With the modeled IF filter, we get, at each discrete spectral value, an SNR according to (15) as follows:

$$\mathrm{SNR}_{\mathrm{IF}} = \mathrm{SNR}\frac{B_s}{B_{\mathrm{ENBW,IF}}}. \tag{15}$$

B_s is the total bandwidth of the measurement system and $B_{\mathrm{ENBW,IF}}$ is the equivalent noise bandwidth [11] of the IF filter. For the modeled IF filter with a bandwidth of 120 kHz, we obtain a $B_{\mathrm{ENBW,IF}}$ of 90.3 kHz. With an IF filter of 120 kHz, the SNR is enhanced by a factor of 44 dB when the input signal only occupies one discrete spectral value. The maximum achievable dynamic range with a single 8-bit ADC with a sampling frequency of 5 GS/s and a modeled IF filter of 120 kHz is approximately 94 dB.

The SFDR achieved with a stationary input signal is limited by spurious originating from nonlinear effects of the ADC. Additionally, the noise floor also depends on the noise figure of the used ADC and preamplifier.

IV. Dynamic Range of a Multiresolution Time-Domain EMI Measurement System

Here, the dynamic performance of the multiresolution time-domain EMI measurement system consisting of several ADCs is investigated. Several ADCs are used in parallel to digitize the analog signal. In the following, an algorithm is described that extracts a floating point value from the discrete values obtained from all ADCs.

A. Floating Point A/D Conversion

In Fig. 4, the intervals that result of a configuration of three ADCs are depicted, which was taken from [4]. Each analog value is simultaneously digitized by all ADCs. The lower and upper boundaries of the amplitude interval i are donated with a_i^l and a_i^h, respectively. If the value is within the range $\left[a_1^l, a_1^h\right]$, the digitized value is extracted from the first ADC. If the value is between $\left[a_2^l, a_1^l[\cup]a_1^h, a_2^h\right]$, the value is taken from the second ADC, and so on. Each extracted value is corrected by the known gain of the ADC. The result is a signal that is digitized as a floating point with a higher dynamic range.

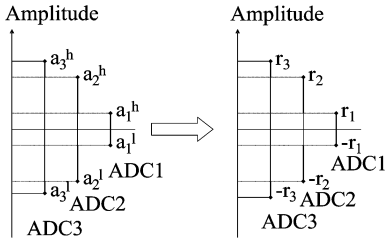


Fig. 4. Intervals for a configuration of three ADCs. From [4].

B. Quantization Noise

In order to maximize the SNR and SFDR, the intervals show in Fig. 4 are to set up to obtain for a given input signal a maximum SNR.

This can be achieved by inserting controlled p-i-n attenuators between the amplifiers and ADCs. In the following, the relationship between the selected amplitude scales for a specific input signal and the SNR are described.

In a practical measurement system by a first measurement with one ADC, the absolute frequency of the digitized values $H[k]$ can be obtained. By the known absolute frequency of digitized values, the optimum parameters are determined. Several measurements can be done to refine the setup.

1) Optimum Amplitude Intervals: A sample that is digitized within the amplitude range $\left[a_1^l, a_1^h\right]$ shows a quantization error that is determined by the quantization interval d_1 of the first ADC. If the sample is within the amplitude range $\left[a_2^l, a_1^l[\cup]a_1^h, a_2^h\right]$, the quantization noise is determined by the quantization step d_2 of the second ADC, and so on. We consider the quantization noise as random error accorded to a midriser quantization and obtain the quantization noise for each ADC according to (7). The absolute frequency of digitized values $H[k]$ can be used to estimate the random noise that is added by a system of several ADCs. We assume that the total noise is a summation of the quantization noise of all ADCs. The quantization noise $P_{N,I}$ of a system consisting of I ADCs is described as

$$P_{N,I} = \frac{1}{12N}\left(d_1^2 \sum_{k=a_1^l}^{a_1^h} H[k] + \sum_{i=2}^{i=I}(d_i^2\alpha_i)\right) \tag{16}$$

with

$$\alpha_i = \left(\sum_{k=a_i^l}^{a_{i-1}^h} H[k] + \sum_{k=a_{i-1}^h}^{a_i^h} H[k]\right). \tag{17}$$

With the used p-i-n attenuators, we can only change the amplification. Any dc offset is eliminated by a high-pass filter. We can assume that $H(k) \approx H(-k)$. We get from (16) and (17),

$$P_{N,I}(r_1, \ldots r_{I-1}) \approx \frac{2}{12N}\sum_{i=1}^{I}\left(d_i^2 \sum_{k=r_{i-1}}^{r_i} H(k)\right) \tag{18}$$

with

$$r_0 = 0. \tag{19}$$

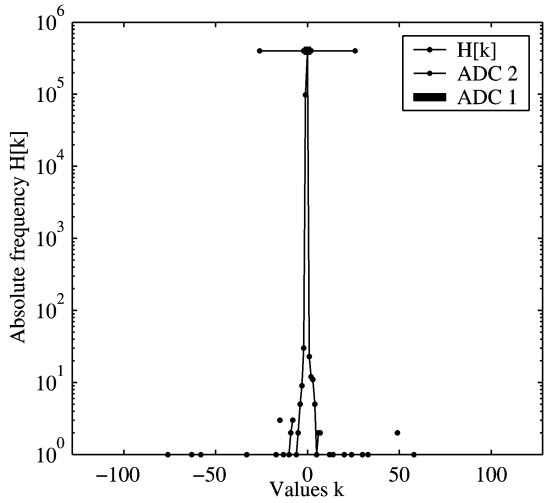


Fig. 5. Optimum amplitude intervals.

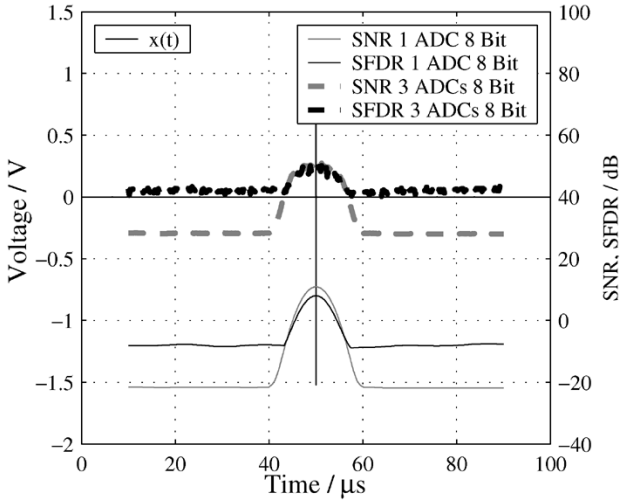


Fig. 6. Estimated instantaneous SNR and dynamic range. Data from [4].

The quantization step d_i of the ADC i can be calculated as follows:

$$d_i = 2^{-B_i} r_i. \tag{20}$$

B_i is the number of bits of the ADC i. In order to maximize the SNR, the quantization noise has to be minimized. By a numerical algorithm, the parameters $r_1, \ldots, r_{I-1}$ are obtained where $P_{N,I}$ shows the absolute minimum.

An example of the parameter obtained for a configuration of three ADCs appears in Fig. 5. The ADC 3 digitizes the full amplitude scale, while the ADC 2 digitizes the range between the values of -26 and $+26$. The interval from -2 to $+2$ is digitized by ADC 1.

2) Accurate Parameter Extraction: During the first measurements, the values were determined by the absolute frequency of quantized values given by a single resolution. Further measurements can be performed to obtain the optimum attenuation levels for the given input signals.

By the multiresolution time-domain EMI measurement system, smaller amplitude intervals, as shown in Fig. 5, can be selected.

C. Dynamic Performance

Here, the dynamic performance of a system consisting of several ADCs is described. We have seen that $x(t)$ has no dc offset. We obtain

$$\langle x(t) \rangle = 0. \tag{21}$$

With (6), we get for the power of the signal

$$P_{\text{Signal}} = \sigma_s^2 = \left\langle \left(x(t)\right)^2 \right\rangle. \tag{22}$$

Applying (6) and (16), we can estimate the instantaneous SNR of the digitized signal. An example is shown in Fig. 6. The data was taken from [4]. It is shown that an improvement of around 40 dB is achieved. In the following, the dynamic parameters are compared with the dynamic parameters that describe the performance of an EMI receiver. We consider the dynamic parameter of the peak detector mode, which describes the worst case. Since the SFDR has been calculated between the signal and an average noise floor, we have to decrease the SFDR by approximately 11 dB to the SFDR for the peak detector mode.

1) SFDR: We obtain from Fig. 6 an estimated SFDR of 39 dB for a multiresolution time-domain EMI measurement system consisting of several 8-bit ADCs.

2) Dynamic Range: The total input range of the system is up to 10 V. The given signal only uses the amplitude range up to 1.2 V. Thus, the dynamic range is 18.4 dB higher than the estimated SFDR. This yields, in this case, to a dynamic range of 57.4 dB.

3) Overload Factor: The overload factor is equal to the given dynamic range. Thus, the estimated overload factor is 57.4 dB.

V. Nonideal Responses

In the following, the nonideal responses of the multiresolution time-domain EMI measurement system are discussed. Methods to reduce those effects by digital signal processing are proposed.

A. Dead Times of Limiters

The dead time τ_d of a limiter is measured by applying a signal that steps from a high amplitude down to zero. The time that is passed while the limiter returns to the linear mode is the dead time.

Typically, p-i-n diodes are used for limiting the input signal. The dead time of a p-i-n diode limiter corresponds to the minority carrier lifetime. The carrier lifetime is determined by the doping level and the material. Typically, p-i-n diodes are doped in a way that the carrier lifetime is between several nanoseconds and several microseconds. p-i-n diodes show the advantage in comparison to Schottky diodes to provide a high breakthrough voltage. For very fast applications, Schottky diodes can be used

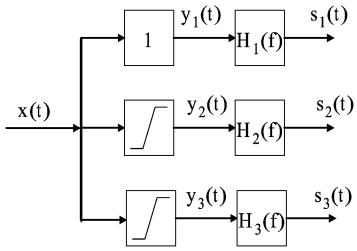


Fig. 7. Frequency response of three channels.

for limiting the signal. The advantage is the short dead time. The dead time is described by (23)

$$\tau_d = \frac{1}{RC}. \tag{23}$$

Typically, the dead time of Schottky limiters is around several nanoseconds. Ultra-fast Schottky diodes show a time constant of several picoseconds, while the breakthrough voltage is usually below 10 V.

B. Frequency Response

Each used ADC shows a frequency response $H_i(f)$. When we use only a single ADC, the frequency response is corrected in the frequency domain after spectral estimation [12]. A system using three channels is depicted in Fig. 7. The input signal is distributed into three channels. The first channel consists of an amplifier with a gain of 1 and has a frequency response of $H_1(f)$. The output signal is according to (24)

$$s_1(t) = x(t) * h_1(t). \tag{24}$$

The second channel has a limiter with the function $l(x)$. We obtain as output signal

$$s_2(t) = l\big(x(t)\big) * h_2(t). \tag{25}$$

$H_2(f)$ shows a behavior in phase and magnitude. A reconstruction by just taking the nonclipped values, as described in Section IV-A, will cause distortion of the signal. The distortion is increased when the different channels also have different frequency responses. The SFDR is degraded. A solution is the correction of the frequency response by digital signal processing before reconstruction.

VI. Hardware Realization

The block diagram of the multiresolution time-domain EMI measurement system has been shown in Fig. 1. Fig. 8 is a photograph of the realized circuit. The whole circuit was implemented as hybrid on a hydrocarbon ceramic substrate. In the following, the various components are investigated.

A. Power Splitter

A resistive power splitter was implemented to obtain a frequency range from dc to 3 GHz. The power splitter was implemented asymmetrically to obtain a low attenuation and a low noise figure at channel 1 where the small signal is digitized. In order to reduce parasitic effects, thick-film resistors were used in the range of 10 Ω up to 270 Ω. The third channel was implemented with a coupling of 22 dB followed by an attenuator of 10 dB. This was done to avoid large resistor values, which results in high parasitics.

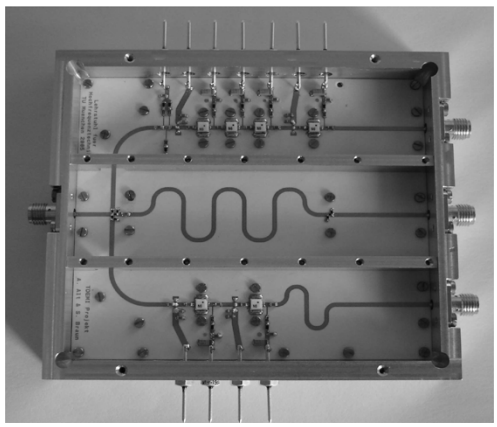


Fig. 8. Realized multiresolution time-domain EMI measurement system.

B. Limiter

1) Conventional Limiter: Conventional limiters [13] use an antiparallel pair of diodes. If a signal passes a certain threshold level, the diodes switch from an off state to the on state. In this state, they show an on resistance $R_{\rm on}$. During the limiting of the signal, the input signal is reflected according to

$$S_{11} = S_{22} = \frac{Z - Z_w}{Z + Z_w} \tag{26}$$

with

$$Z = \frac{Z_w R_{\rm on}}{Z_w + R_{\rm on}}. \tag{27}$$

For a typical Schottky diode with a $R_{\rm on}$ of 7 Ω and a characteristic impedance of 50 Ω, we obtain a reflection coefficient of −2.14 dB. In this mode, we obtain a maximum attenuation of the input signal of 13.2 dB. A pulse with an amplitude of 6 V would be reflected with 4.7 V. The limiting value would be 1.3 V.

2) Limiter With Improved Matching: The multiresolution time-domain EMI measurement system uses a novel limiter circuit with improved matching. The low reflection coefficient during the limiting of the input signal is necessary to avoid any reflections that will contribute to measurement errors on the other channels. The circuit of the limiter with improved matching is shown in Fig. 9. A combination of an antiparallel pair and a serious pair of Schottky diodes are used. The bridged resistor R_b improves the matching. When a small signal is connected to the input, the limiter operates in the linear operation mode. Due to the forward resistance of the diodes that are pre-biased by the voltage source V_s, they show an on resistance of around 7 Ω. The attenuation during this state is approximately 2 dB. When a large signal is applied to the input, the series diodes D_{S1} or D_{S2} switch to an high impedance

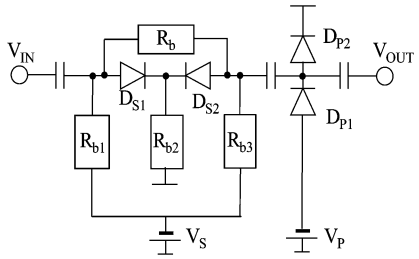


Fig. 9. Diagram of the limiter circuit.

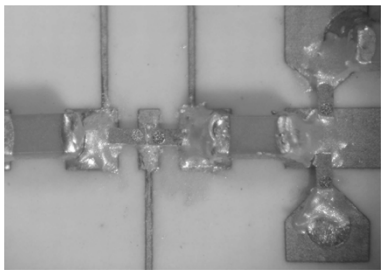


Fig. 10. Realized limiter circuit without the bridged resistor R_b.

state. The diodes D_{P1} or D_{P2} switch to an on state. For a given Z_w, the bridged resistor should have a resistance of

$$R_b = Z_w - R_{\rm on}. \tag{28}$$

Due to the resistor, the input impedance is matched. Ultra-fast Schottky diodes have been selected to reduce dead times. To avoid ringing effects, packages with very low parasitic inductances have to be used. We selected a GaAs Schottky diode with a capacitance of 60 fF and a parasitic inductance below 0.1 nH. A photograph of the realized limiter circuit is shown in Fig. 10. The Schottky diodes were fixed with a conductive epoxy glue on the substrate, while the rest of the components were conventionally soldered. The bridged resistor is placed above the diodes between the two capacitors. This is done to minimize parasitics.

3) Dynamic Range: The maximum input voltage of the limiter circuit depends on the used diodes. For transients with low energy and high amplitude, the maximum input voltage is described by (29)

$$V_{\rm max} = V_{\rm br}\frac{R_b + R_{\rm on}}{R_b}. \tag{29}$$

$V_{\rm br}$ describes the reverse breakthrough voltage of the diode. For a stationary signal, the power that is dissipated by both diodes limits the maximum input power. The dissipated power of the diodes are given by (30)

$$P_{\rm diss} = \frac{V_{\rm out}^2}{R_{\rm on}}. \tag{30}$$

Concerning $R_{\rm on} \ll Z_w$, we obtain

$$P_{\rm diss} = \frac{\left(V_{\rm in}\dfrac{R_{\rm on}}{R_b + R_{\rm on}}\right)^2}{R_{\rm on}}. \tag{31}$$

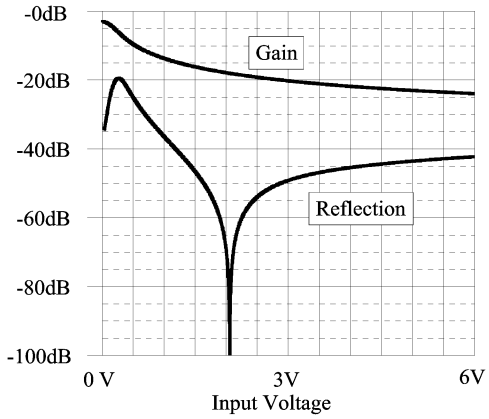


Fig. 11. Simulated attenuation and reflection in dependence of the input voltage.

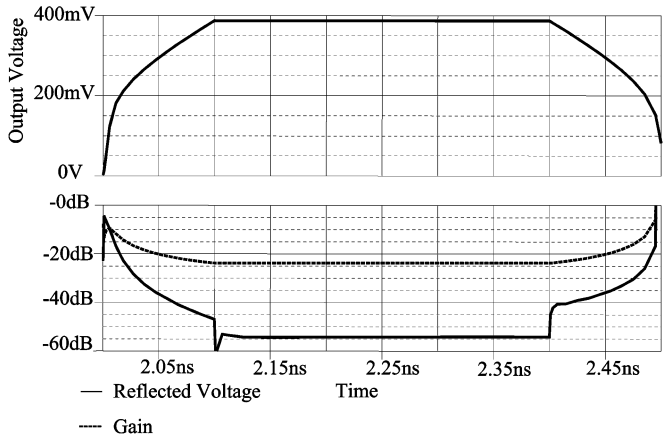


Fig. 12. Simulated response of the limiter to a 6-V 0.3-ns pulse.

We suppose that the power is dissipated equally in both diodes. We obtain the total maximum input power

$$P_{\rm max} = 2\frac{P_{\rm diss}(R_b + R_{\rm on})^2}{Z_w R_{\rm on}}. \tag{32}$$

The used Schottky diodes have a $V_{\rm br}$ of 5 V and will retain a maximum power dissipation $P_{\rm diss}$ of 75 mW. The maximum input amplitude for short transients is 5.8 V. The maximum continuous wave input power is 1 W, which could be thermally absorbed. However, due to the limitations by the breakthrough voltage, the input power is limited to 250 mW, which corresponds to 24 dBm. Though a voltage above the breakthrough voltage will not destroy the diode because the current is limited, it will contribute a mismatch of the impedance. Thus, at the limiter circuit, a maximum of 5.8 V at the input should be attached. Typically, an EMI receiver has a maximum input range up to 15 dBm when the maximum attenuation level of 60 dB is selected.

4) Simulations: Time-domain simulations where performed with SPICE using the diode model of the used diode. The result is shown in Fig. 11. The reflected peak voltage is attenuated by 42 dB. The response to a 0.3-ns pulse is shown in Fig. 12. The pulse has an amplitude of 6 V. It is shown that the pulse is limited to 400 mV with a return loss of over 40 dB.

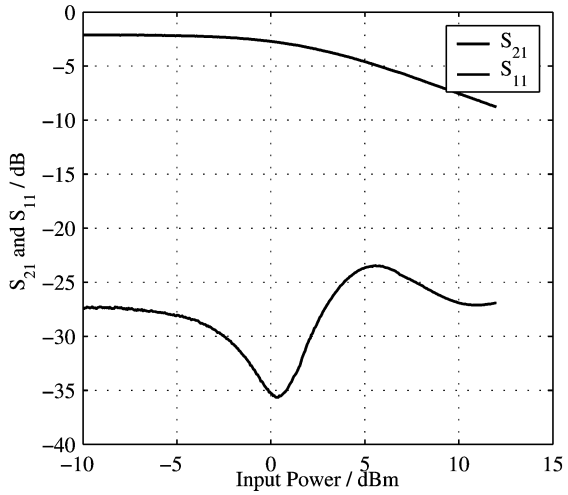


Fig. 13. Response of the limiter circuit in a power sweep.

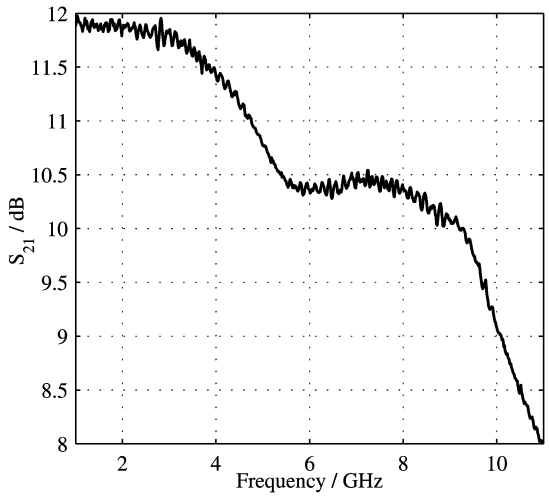


Fig. 14. Frequency response of a single amplifier stage.

5) Measurements: A measurement with a vector network analyzer using a power sweep has been carried out. The input power was swept from −10 to 12 dBm. The measurement was performed at 1 GHz. The results are shown in Fig. 13. The measurements show that an attenuation of 9 dB with a reflection coefficient of −27 dB have been achieved. The attenuation during the linear operation mode is 2.0 dB.

C. Amplifier

In order to get a broad-band maximum flat frequency response up to several gigahertz, a cascadable ultra-broad-band monolithic-microwave integrated-circuit (MMIC) GaAs amplifier was used for an operating range from 30 kHz to 11 GHz.

A measurement with a vector network analyzer of a single amplifiers was carried out. The result is shown in Fig. 14. The amplifier shows a 3-dB bandwidth of 10 GHz.

D. Digitally Controlled p-i-n Attenuator

In order to set each channel to a selectable amplitude scale, p-i-n attenuators are used. These p-i-n attenuators use digital controlled current sources to control the bias currents of the p-i-n diodes. The currents for an optimum input impedance and maximum flat frequency response were experimentally investigated. The currents are stored in an lookup table, which is used to set up the proper current for a selected attenuation. p-i-n attenuators have been designed for a frequency range from 30 MHz to 3 GHz.

TABLE I
NOISE BEHAVIOR OF THE MULTIRESOLUTION TIME-DOMAIN EMI MEASUREMENT SYSTEM

CH	NF	Thermal Noise	Quantization Noise	Amplitude
1	9.2	-165 dBm/Hz	-172 dBm/Hz	1.8 mV
2	27.2	-147 dBm/Hz	-131 dBm/Hz	200 mV
3	35.5	-138 dBm/Hz	-97 dBm/Hz	10 V

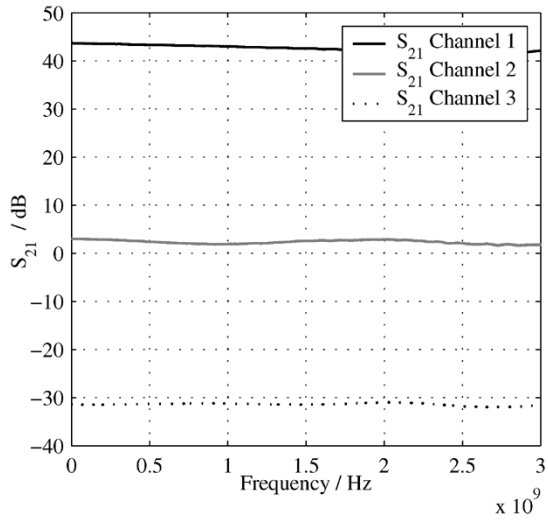


Fig. 15. Frequency response channels 1–3.

E. Noise Figure

The noise floor of the low-noise multiresolution time-domain EMI measurement system is calculated with the known noise figure of all pre-stages. In order to calculate the spectral noise density at the input of each ADC, we calculate the noise figure of each channel according to (33)

$$F = 1 + F_1 + \frac{F2}{G1} + \frac{F3}{G1G2} + \cdots + \frac{F_N}{\prod_{k=1}^{N-1} G_k}. \qquad (33)$$

In Table I, an overview of the different channels are shown. The quantization noise is given for an 8-bit ADC with 5 GS/s, where we take into account that we have used a modeled IF filter of 120 kHz, which shows an equivalent noise bandwidth $B_{\mathrm{ENBW,IF}}$ of 90.3 kHz. We would obtain an average noise floor of $-8.4\ \mathrm{dB} \cdot \mu\mathrm{V}$.

F. Frequency Response

The frequency response of the implemented circuit was measured with a VNA. The result is shown in Fig. 15.

VII. Measurement Results

In order to investigate the SNR and SFDR, a pulse generator with a pulsewidth of 500 ps was connected via a notch filter with an attenuation of at least 40 dB at 300 and 900 MHz to the multiresolution time-domain EMI measurement system without

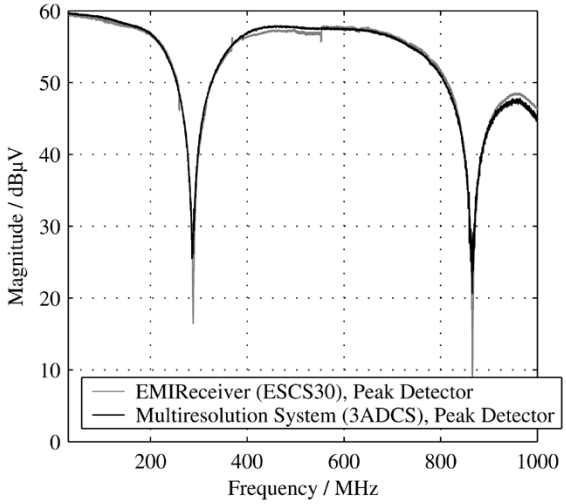


Fig. 16. Measurement of 0.5-ns pulse in the peak detector mode.

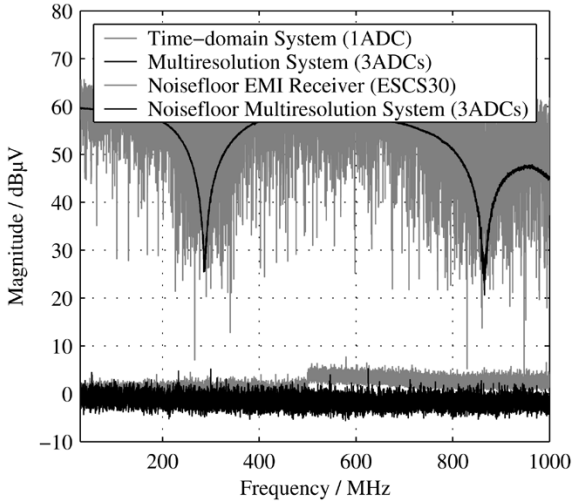


Fig. 17. Measurement of 0.5-ns pulse and noise floor in the peak detector mode.

using the optional p-i-n attenuators. This test is described by CISPR 16-1 [2] to determine the SFDR for transient signals. A measurement was carried out with the multiresolution time-domain EMI measurement system and compared with results obtained with a conventional full compliance EMI receiver. It is shown that the multiresolution time-domain EMI measurement system shows an SFDR of 34.5 dB. The conventional EMI receiver shows an SFDR of 43.5 dB. The maximum difference between both measurements is over the complete frequency range of 1 dB. The result is shown in Fig. 16. The used pre-selection filters and the stepped attenuators of the EMI receiver show, in this case, measurement errors that are below 2 dB. The largest measurement error is caused at 550 MHz. At this frequency, the used EMI receiver ESCS 30 from Rohde&Schwarz switches between two channels of pre-selective filter sections.

In order to investigate the dynamic range, a measurement has been carried out, which compares the noise floor of the EMI receiver ESCS 30 with the noise floor of the multiresolution time-domain EMI measurement system. A measurement was also performed with the pulse generator and notch filter. The results of the multiresolution time-domain EMI measurement system with three ADCs are compared with results obtained with the time-domain EMI measurement system operating with one ADC. The results are shown in Fig. 17. It is shown that the multiresolution time-domain EMI measurement system with three ADCs in comparison to the time-domain EMI measurement system shows an improvement of the dynamic range of approximately 60 dB. The SFDR is enhanced by approximately 30 dB. The noise floor of the system is below the noise floor of a conventional EMI receiver. A system must have a 22-bit ADC to obtain the same dynamic range without pre-selection or selectable attenuation. If a selectable attenuator is used, at least a 16-bit ADC must be used to obtain the same SFDR.

The overload factor of the different measurement systems have been investigated. The demodulated IF signal of a conventional EMI receiver was compared with the extracted IF signal of the multiresolution time-domain EMI measurement system with three ADCs and compared with results of the time-domain EMI measurement system, which uses one ADC.

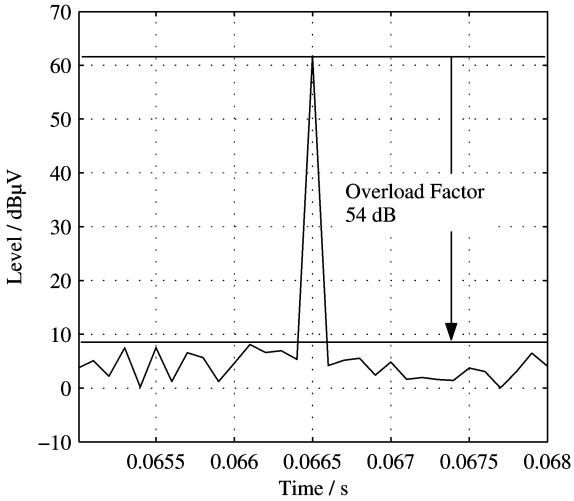


Fig. 18. Overload factor of the conventional EMI receiver.

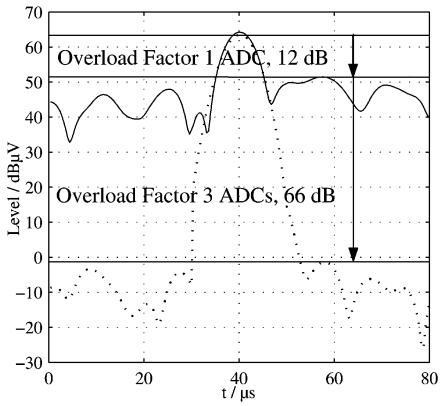


Fig. 19. Overload factor of the multiresolution system with three ADCs and the time-domain EMI measurement system with one ADC.

The measurement with the EMI receiver is shown in Fig. 18. The EMI receiver ESCS 30 shows a typical overload factor of 53 dB. The measurement performed with the time-domain EMI measurement system and the multiresolution time-domain EMI measurement system is shown in Fig. 19 The multiresolution time-domain EMI measurement system has an overload factor

of 66 dB. This is an improvement in comparison to the EMI receiver of 13 dB and an improvement of 54 dB in comparison to a system with one ADC.

VIII. Conclusion

A low-noise multiresolution high dynamic ultra-broad-band time-domain EMI measurement system has been presented. The system has shown a lower SFDR than conventional EMI receivers. For stationary signals, the system fulfills the standards according to the SFDR. For a transient signal, the system still has a SFDR that is, in the worst case, 1.5 dB below the requirements of CISPR 16-1. An improvement of 30 dB in comparison to a system with one ADC has been shown. The presented multiresolution time-domain EMI measurement system has a 5-dB lower noise floor than EMI receivers. The multiresolution time-domain EMI measurement system can measure stationary signals with a higher input power than conventional EMI receivers with internal 60-dB attenuator. It was shown that the multiresolution time-domain EMI measurement system has a 12-dB higher overload factor than conventional EMI receivers that uses stepped attenuators and a preselection filter. Due to no preselection, the input VSWR is better than 1.3 for the frequency range of 30 kHz–3 GHz independent for an input power range up to 24 dBm. Measurements performed with the multiresolution time-domain EMI measurement system are 50 times shorter than measurements performed with an EMI receiver.

IX. Outlook

With the availability of high-speed ADCs of several gigasamples per second in the future, time-domain EMI measurements with a high dynamic range and high SNR will be possible for the frequency range of several gigahertz without any preselection. By using Schottky diodes with a lower resistance in the on state, the noise figure of the system can still be decreased. For higher frequency ranges, the resistive power splitter can be replaced by any asymmetric power splitter with a lower noise figure.

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